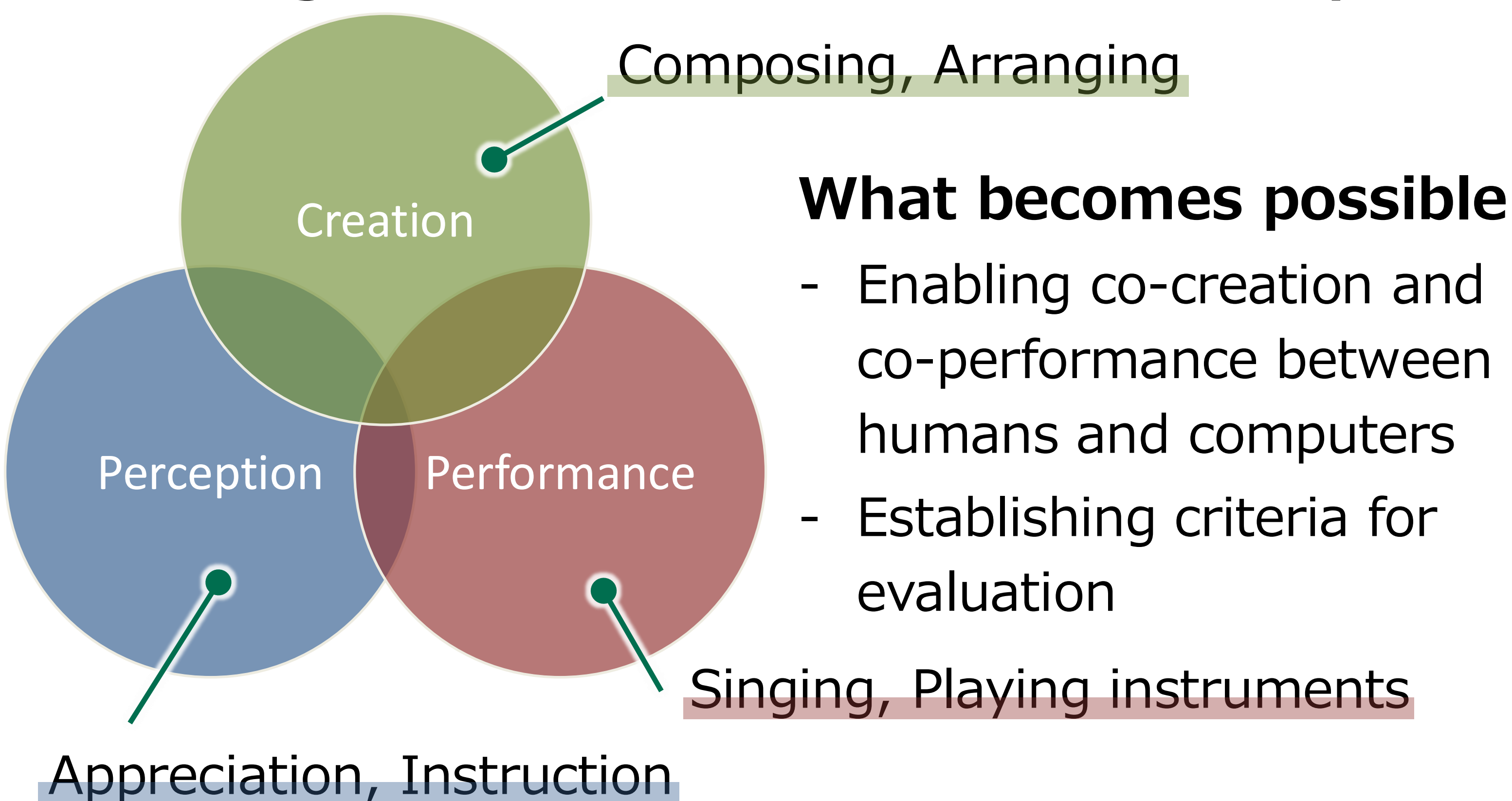


Modeling Human Music Activities : A Hierarchical Symbolic Music Generation Model with Chord Controllability

Keito Sawada Toda Lab. D2

Doctoral Research Goal

Modeling human music activities on a computer



Creation

Modeling human music creation

- Improve generation of global musical structure
- Enable controllable automatic arrangement

Contributions

- Understanding structural patterns in human-composed music
- A generation model that captures both global and local structure
- Exploration of new musical styles

Perception

Modeling human music perception

- Address the lack of objective evaluation metrics
- Incorporate human sensitivity into evaluation

Contributions

- Understanding mechanisms of music perception
- Development of new evaluation metrics
- Supporting music education and feedback

Performance

Modeling human performance and instruction

- Accurate note timing at fixed tempo is largely solved
- Expressive elements beyond pitch and rhythm remain challenging
- Toward feedback including timing deviations and dynamics

Contributions

- Understanding how humans interpret musical scores
- Supporting practice independent of time and location

Creation – Symbolic Music Generation

Problem settings

- Generating music in a symbolic form such as MIDI
- Conditioned by chord progression information

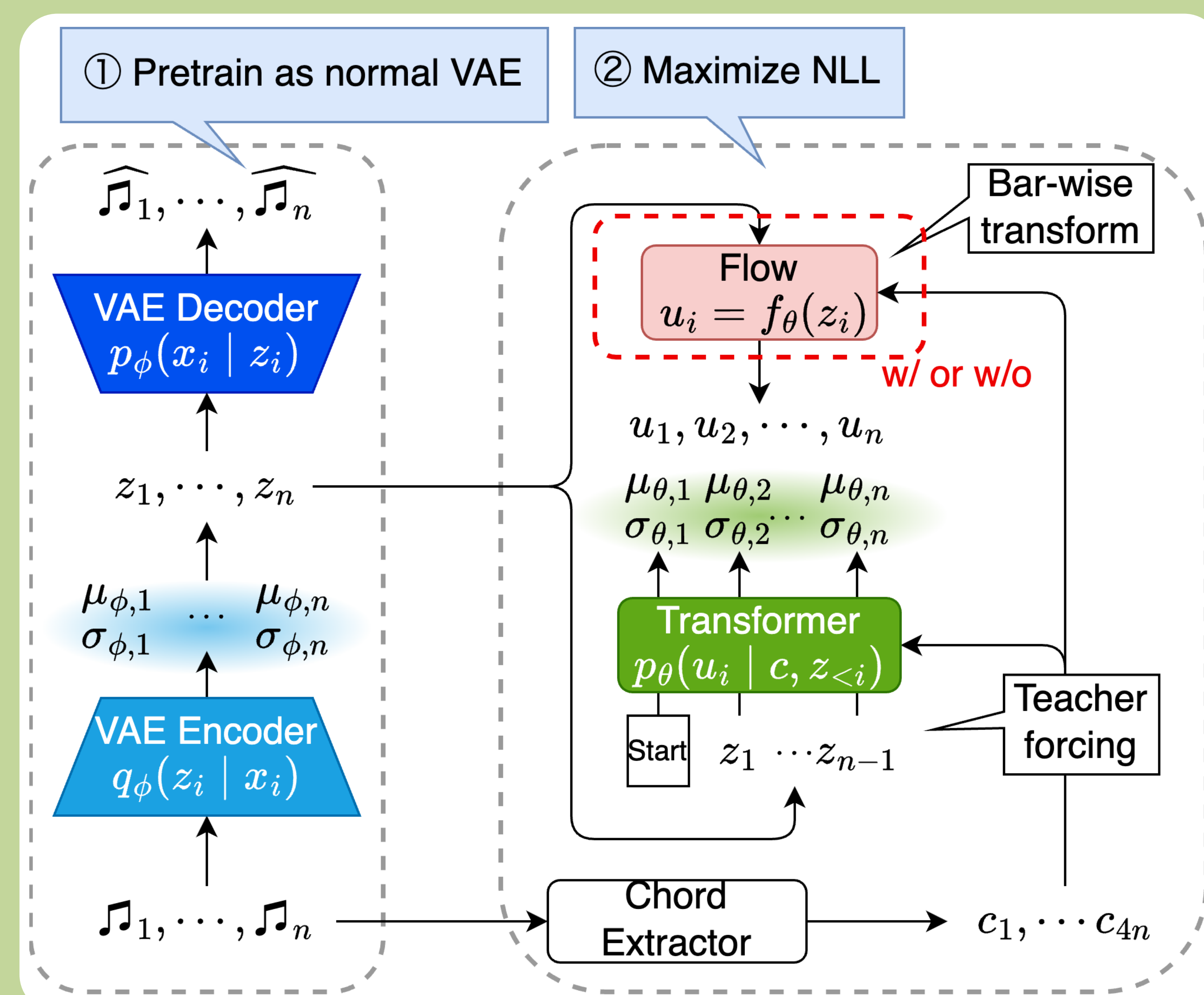
Motivation

- Music has global structure across bars and phrases
- I use hierarchical modeling to efficiently capture this long-term structure with bar-wise feature
- Can Flow enhances expressiveness for better local quality and controllability?

Proposed method

Step 1: Mapping between bar-wise music and feature

Step 2: Learning bar-wise feature sequence



Metrics

- D_P, D_D [1] : Evaluates tendency of notes property
- CD [1] : Evaluates chord controllability
- BFS [2] : Evaluates global musical structure

Objective Evaluation Results

Length	method	$D_P \uparrow$	$D_D \uparrow$	CD $\downarrow$	BFS $\downarrow$
8 bars	Polyffusion [3]	0.896	0.865	1.148	3.33
	Ours w/ Flow	0.787	0.729	0.476	30.583
	Ours w/o Flow	0.877	0.874	0.716	3.43
32 bars	Polyffusion [3]	0.853	0.943	1.670	4.23
	Ours w/ Flow	0.774	0.751	0.733	44.488
	Ours w/o Flow	0.802	0.802	0.717	2.39

- w/o Flow improved global structure in longer music
- w/ Flow has better chord controllability in any length
- Both methods need improvement in local structure

Conclusion

- Proposed hierarchical method improved chord controllability and global structure in long generation
- Local structure needs further improvement

Reference

- [1] Y. Ren, et. al., "Popmag: Pop music accompaniment generation"
- [2] K. Sawada, et. al., "Hierarchical Symbolic Music Generation with Variational Autoencoder-based Bar-wise Feature Sequence"
- [3] L. Min, et. al., "Polyffusion: A Diffusion Model for Polyphonic Score Generation with Internal and External Controls"